

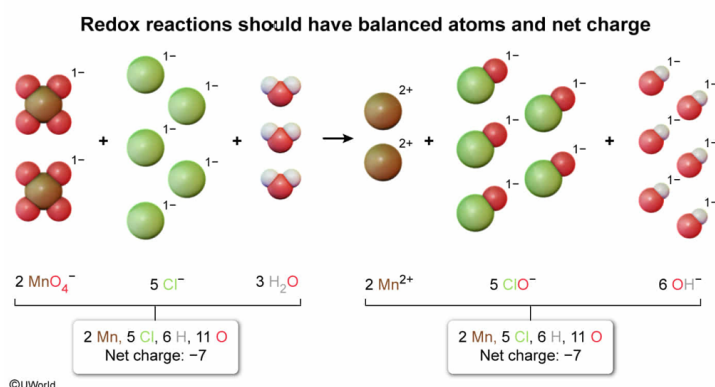
If  $\text{MnO}_4^-(aq)$  undergoes an oxidation-reduction reaction with  $\text{Cl}^-(aq)$  in an aqueous solution at pH 10, which of the following gives a balanced chemical equation for the reaction?

- ☐ A.  $\text{MnO}_4^-(aq) + 3 \text{Cl}^-(aq) + 5 \text{H}_2\text{O}(l) \rightarrow \text{Mn}^{2+}(aq) + 3 \text{HClO}(aq) + 6 \text{OH}^-(aq)$  [12%]
- ☐ B.  $\text{MnO}_4^-(aq) + 3 \text{Cl}^-(aq) + \text{H}_2\text{O}(l) \rightarrow \text{Mn}^{2+}(aq) + 3 \text{ClO}^-(aq) + 2 \text{OH}^-(aq)$  [29%]
- ✓ ☒ C.  $2 \text{MnO}_4^-(aq) + 5 \text{Cl}^-(aq) + 3 \text{H}_2\text{O}(l) \rightarrow 2 \text{Mn}^{2+}(aq) + 5 \text{ClO}^-(aq) + 6 \text{OH}^-(aq)$  [47%]
- ☐ D.  $2 \text{MnO}_4^-(aq) + 5 \text{Cl}^-(aq) + 11 \text{H}^+(aq) \rightarrow 2 \text{Mn}^{2+}(aq) + 5 \text{HClO}(aq) + 3 \text{H}_2\text{O}(l)$  [10%]

Correct

47% answered correctly

## Explanation:



**Oxidation-reduction (redox) reactions** consist of two simultaneous **half-reactions** (one for the **oxidation** and one for the **reduction**) that add together to give the overall reaction. In a balanced redox reaction, the total number of electrons released from the oxidation half-reaction must equal the total number of electrons used by the reduction half-reaction. Consequently, the first step in balancing a redox reaction is to balance the electrons exchanged.

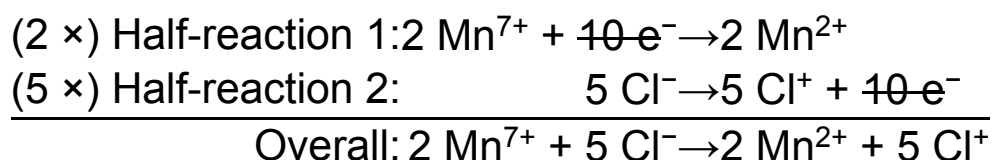
In the given reaction, the **Mn atoms are reduced** from an **oxidation state** of +7 in  $\text{MnO}_4^-(aq)$  to a state of +2 as  $\text{Mn}^{2+}(aq)$  ions, a gain of five electrons per Mn atom. As such, the reduction half-reaction (excluding the  $\text{O}^{2-}$  components from the chemical formula) can be written as:



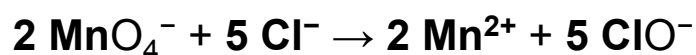
Simultaneously, **Cl atoms are oxidized** from an **oxidation state** of -1 as  $\text{Cl}^-(aq)$  ions to a state of +1 in the  $\text{ClO}^-(aq)$  ions, a loss of two electrons per Cl atom. As such, the oxidation half-reaction (excluding the  $\text{O}^{2-}$  component from the chemical formula) can be written as:



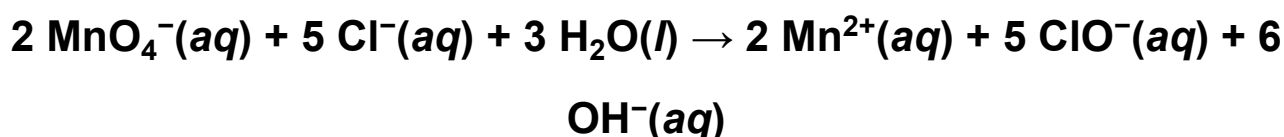
To balance and **cancel the electron terms** in the **overall reaction**, **multiples** of both half-reactions can be added together.



Based on the electron balance above, the coefficients of the Mn and Cl species in the balanced reaction must be:



The next step is to **balance the atoms** and the **net charge**. Because the reaction occurs in an **aqueous solution** at pH 10 (ie, under **basic conditions**),  $\text{H}_2\text{O}$  and  $\text{OH}^-$  can be added as reactants or products to balance the equation. Therefore, the fully balanced redox reaction is:



**(Choice A)** The atoms are not balanced in this equation (ie, 10 H atoms in the reactants but 9 H atoms in the products).

**(Choice B)** The net charge is not balanced in this equation (ie,  $-4$  for the reactants but  $-3$  for the products).

**(Choice D)** This equation indicates acidic conditions (ie,  $H^+$  is a reactant). The question states that the given reaction occurs under *basic conditions* (ie, pH 10).

**Educational objective:**

Redox reactions are fully balanced when the same number of each type of atom and the same net charge are present on both sides of the reaction equation.

**Time spent:**0 sec

**Subject:**  
General Chemistry

**QID:**403903

**System:**  
Interactions of Chemical Substances